Best Practice: Simulation II
Scientific Methods and Writing

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Plan for today (90 min)

Min	Part
30	A. Warm-up questions
5	B. Roadmap: from results to paper
25	C. Live demo: Jupyter notebook
10	D. Repository and virtual environment
15	E. From results to paper: figures, structure, writing
3	F. Outlook: ML model for the Hardening Soil model
2	Take-home message

Part A: What do you already know?

- Twelve short questions.
- Just answer them.
- We explain only where the answer is weak.

Q1: Where do you search for papers?

Which tools do you use?

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Which tools do you use?

- Google Scholar.
- Scopus and Web of Science.
- TU Graz library.
- ResearchGate (authors share their papers).
- Follow the references of a good paper.

Q2: Is the source reliable?

How do you judge a source?

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How do you judge a source?

- Peer review.
- Authors and institution.
- Citations.
- Data and code available.

Q3: How do you read a paper?

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1. Title and abstract.
2. Figures and tables.
3. Conclusion.
4. Only then: the details.

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Try it with our paper (open access):



doi:10.1007/s11440-026-02951-5

Q4: Reference manager

What is a reference manager?

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What is a reference manager?

- Software that stores and formats your citations.
- Examples: Zotero, Citavi, Mendeley.
- **Recommendation:** Citavi is mostly used at our institute. Other options are also generally recommended.

Q5: Structure of a paper or thesis

What are the usual parts of a paper or thesis?

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Research-oriented

1. Introduction.
2. Background.
3. Methods.
4. Results.
5. Discussion.
6. Conclusion and outlook.

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What are the usual parts of a paper or thesis?

Research-oriented

1. Introduction.
2. Background.
3. Methods.
4. Results.
5. Discussion.
6. Conclusion and outlook.

Application-oriented

1. Introduction.
2. Project description.
3. Soil conditions.
4. Numerical modelling.
5. Analysis.
6. Discussion.
7. Conclusion.

Q6: Where is the research question?

In which part do you find the research question?

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In which part do you find the research question?

- In the introduction.
- The conclusion **must** answer it.

Q7: How do you start writing?

Where do you begin: with the introduction?

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Where do you begin: with the introduction?

1. Make an outline: the headings first.
2. Think about distribution in percentages.
3. Start with figures, tables and results (not necessarily in final quality).
4. Write the methods while the work is fresh!
5. Write the introduction and the abstract last.
6. A first draft may be rough. Improve it later.

Q8: How do you build a paragraph?

What is the structure of a good paragraph?

Q8: How do you build a paragraph?

What is the structure of a good paragraph?

1. Topic sentence: it states the main point (often first sentence of a paragraph).
2. One idea only, with evidence or explanation.
3. Short summary or link to the next paragraph, if needed.

Common fault: one paragraph with several ideas or a number of topic sentences.

Q9: How do you present a figure?

What do you write about a figure?

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What do you write about a figure?

1. Say what the figure shows: axes, units, curves.
2. Describe what you see, one item after the other!
3. **Explain the results**.
4. Only then draw the conclusion.

Common fault: a conclusion without a description.

Q10: How do you write the method?

How can another person repeat your work?

Q10: How do you write the method?

How can another person repeat your work?

- Describe every step, in the order you did it.
- Give all inputs: data, parameters, software versions, hardware specs.
- Share the code and the data.

Common fault: missing steps and missing numbers.

Q11: Reproducibility

What is a reproducible result?

Q11: Reproducibility

What is a reproducible result?

- Another person gets the same result.
- They use your code and your data.
- $\rightarrow$ This is the topic of the next part of the lecture.

Q12: Programming

Who has used Python or Jupyter before?

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Who has used Python or Jupyter before?

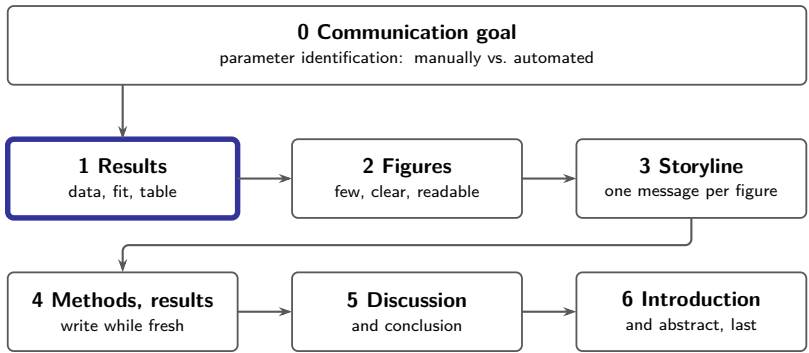
- If not, don't worry. We'll cover the basics.
- If yes, you can help your neighbor.
- Enjoy the live demo. Please ask questions.

Part B: From results to paper. The plan

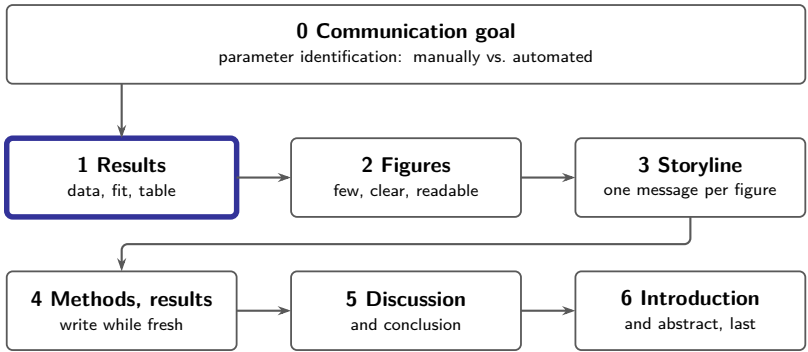
0 Communication goal

parameter identification: manually vs. automated

Part B: From results to paper. The plan



Part B: From results to paper. The plan



In the demo we make **stage 1**: the results. After the repository part we go through **stages 2 to 6**.

Part C: Workflow of the demo

Goal: find the soil parameters from a noisy lab test.

1. Model: parameters $\rightarrow$ curve.
2. Measurement: noisy data.
3. Guess the parameters by hand.
4. Let an optimization algorithm search.
5. Reveal the true values and compare (evaluation).
6. Save data and results.

Note: This is a simplified example for educational purposes.

Step 1: The model maps the parameters to a curve

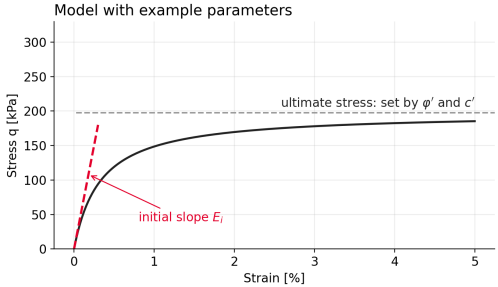
Duncan Chang Model:

Symbol	Meaning
E_i	Initial stiffness [kPa]
φ'	Friction angle [deg]
c'	Cohesion [kPa]

Fixed: failure ratio $R_f = 0.9$, cell pressure $\sigma'_3 = 100$ kPa.

$$q = \frac{\varepsilon_a}{\frac{1}{E_i} + \frac{\varepsilon_a}{q_{ult}}}$$

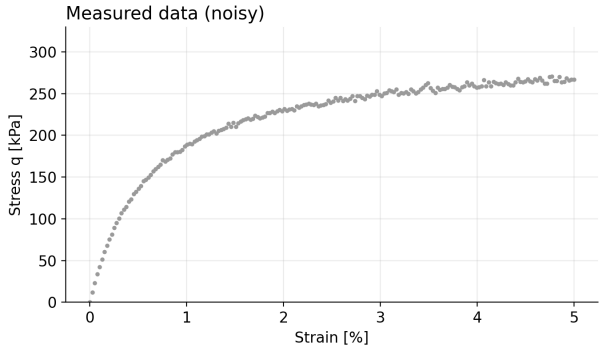
$$q_{ult} = \frac{q_f}{R_f}, \qquad q_f = \frac{2c' \cos\varphi' + 2\sigma'_3 \sin\varphi'}{1 - \sin\varphi'}$$



Example values.

Step 2: The measurement

- A lab test gives noisy data.
- Real tests are never perfect.
- The true parameters are hidden.
- Task: find the parameters that explain these data.



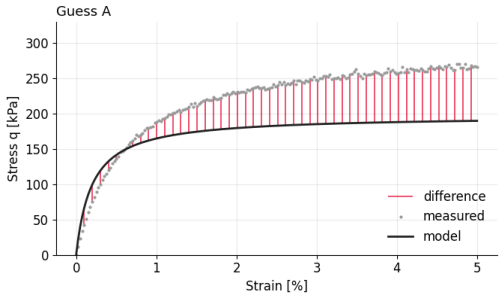
Step 3: Your turn. Guess the parameters

1. Each student (or groups of 2 students) gives three numbers: E_i, φ', c'.
2. We type them into the notebook (step 3b).
3. The notebook plots all curves and ranks the RMSE.
4. Who is closest to the data?

Bounds: E_i 10 000–200 000 kPa, φ' 20–40°, c' 0–30 kPa.

How good is a guess? The RMSE

- RMSE is one number for the misfit between model and data.
- Red lines: the difference at each strain point.
- Small RMSE means a good fit.



RMSE = 55.5 kPa

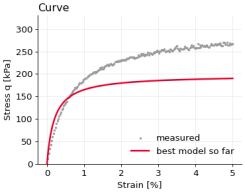
Parameter	Guess A
E_i [kPa]	100 000.0
phi [deg]	25.0
c [kPa]	10.0

Step 4: Let the computer search. Three lines of code

The algorithm changes the parameters step by step until the RMSE is smallest.

```
bounds = [(10e3, 200e3), (20, 40), (0, 30)]
start  = [100e3, 25, 10]
result = calibrate(strain, q_meas, start, bounds)
```

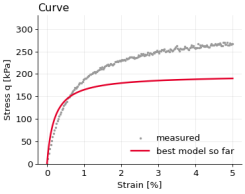
Step 4: The search. The curve and the parameters

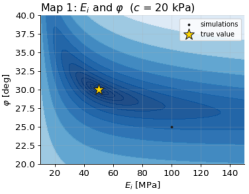


RMSE = 55.5 kPa

Parameter	True	Best so far
E_i [kPa]	50 000.0	100 000.0
phi [deg]	30.0	25.0
c [kPa]	20.0	10.0

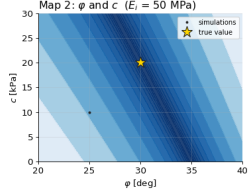
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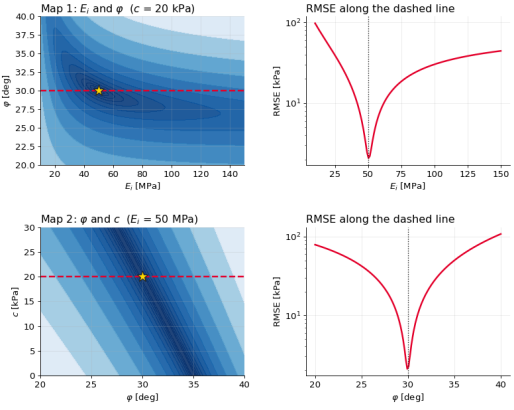


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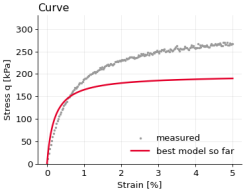
Parameter	True	Best so far
E_i [kPa]	50 000.0	100 000.0
phi [deg]	30.0	25.0
c [kPa]	20.0	10.0

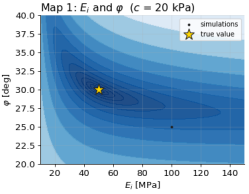


Step 4: Reading the maps



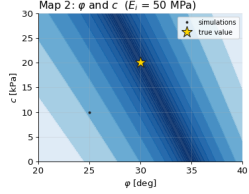
Step 4: The search in action



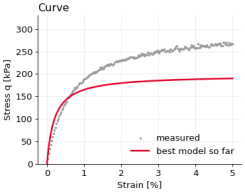


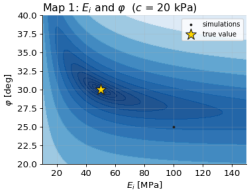
RMSE = 55.5 kPa

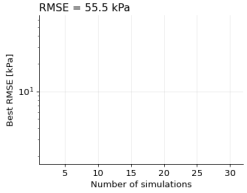
Parameter	True	Best so far
E_i [kPa]	50 000.0	100 000.0
phi [deg]	30.0	25.0
c [kPa]	20.0	10.0



Step 4: The search in action. Evaluation

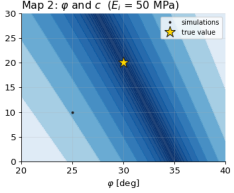


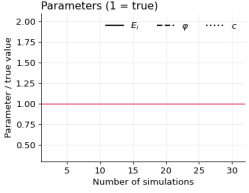




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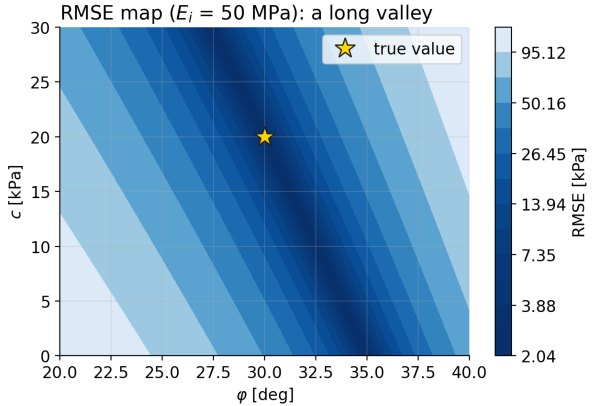
Step 5: Result

Parameter	True	Fitted
E_i [kPa]	50 000	50 040
φ' [deg]	30.0	30.6
c' [kPa]	20.0	17.6

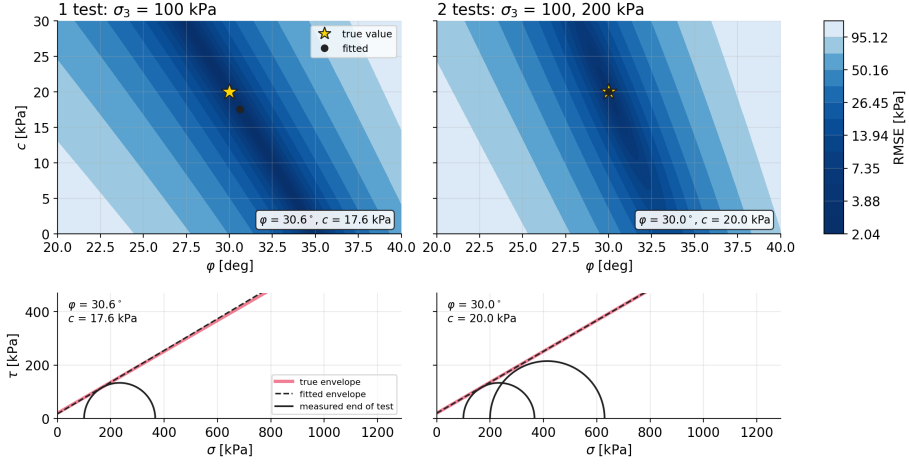
- E_i is found well.
- φ' and c' are less certain.
- Both change the curve in a similar way.
- This is the *identifiability* problem.

Why are φ' and c' less certain?

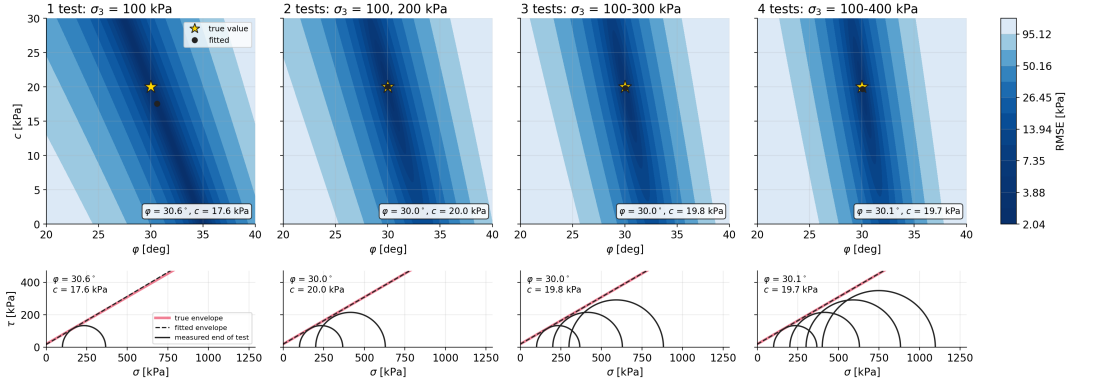
- The RMSE map has a long, narrow valley.
- Many pairs of φ' and c' give almost the same curve.
- Noise moves the minimum along the valley.
- More tests at other σ_3' make the valley shorter.



More tests make the valley shorter



More tests make the valley shorter



Step 6: Save the data and the results

- The measurement is in `data/`.
- The result table goes to `results/`. The figures go to `graphics/`.
- Next: keep all files in one repository.

```
table.to_csv('../results/parameters.csv')
```

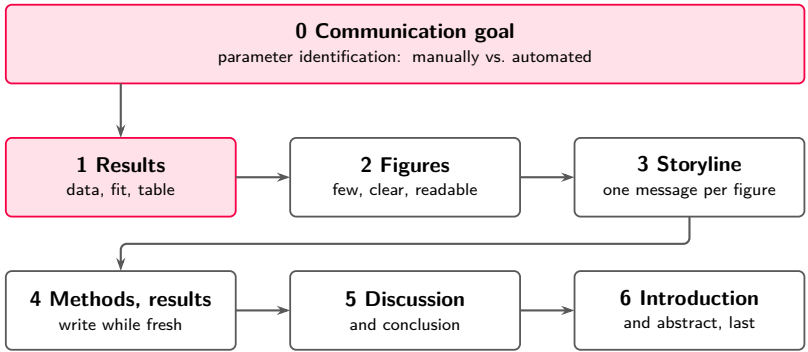
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```
table.to_csv('../results/parameters.csv')
```

- **Important:** Clear structure is necessary for reproducibility. You will have to deal very often with a lot of data.

Part D: Keep the results safe



Stage 1 is done: we have results. Now we keep data, code and results in one repository. It helps in every later stage!

Part D: Folder structure

```
duncan-chang-calibration/
|-- data/               measurement and true values
|-- src/                model, helpers, figure scripts
|-- notebooks/          01_calibration_basics.ipynb
|-- results/            output tables
|-- graphics/           figures, GIFs, QR codes
|-- slides/             lecture slides (LaTeX)
|-- tests/              automatic checks
|-- docs/               lecture plan
|-- .gitignore          files that Git ignores
|-- requirements.txt    list of Python packages
|-- README.md           start here
```

Why a virtual environment?

- A virtual environment is a private set of Python packages.
- Each project has its own versions.
- Other projects can use the same environment (can work on other computers).
- Other people can build the same setup.
- `requirements.txt` lists the packages.

Create the environment (Windows)

```
python -m venv .venv
.venv\Scripts\activate
pip install -r requirements.txt
```

- Open `notebooks/01_calibration_basics.ipynb` in VS Code.
- Choose the kernel from `.venv`.
- PowerShell can block the activation. Use the Command Prompt then.

Save the work with Git: the commands

```
cd duncan-chang-calibration          # go into the project folder
git init                             # start tracking this folder
git add .                            # select all files
git commit -m "First␣version"        # save a snapshot
git branch -M main                   # name the main branch "main"
git remote add origin <repo-url>     # tell Git where the online copy is
git push -u origin main              # send the snapshot to GitHub
```

- A *remote* is the copy of the repository on GitHub. `origin` is its nickname.
- Without a remote, the project stays only on your computer.
- GitHub Desktop sets the remote for you when you press *Publish repository*.

Publish with GitHub Desktop

1. File → Add local repository: choose the project folder.
2. Write a short summary. Press *Commit to main*.
3. Press *Publish repository*.
4. Remove the tick at *Keep this code private*.
5. Check on github.com: the files are online.

Before you publish

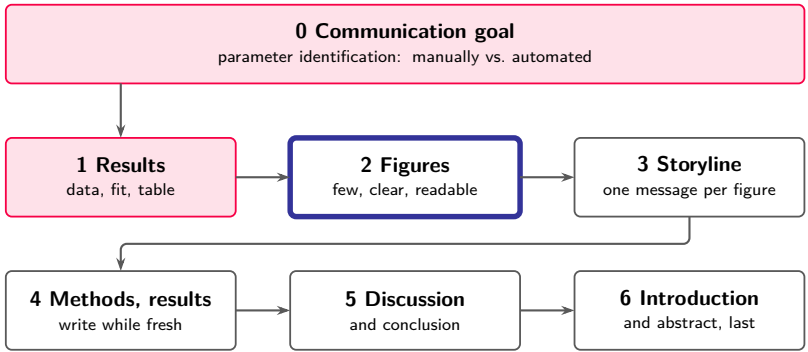
No tokens, no passwords, no `.venv` folder.

Benefits of a repository

- Backup of all files.
- History of every change.
- Easy sharing with the supervisor, community, reviewers, parents, friends, ...
- Citable with a DOI (Zenodo).
- Reproducible for the reader.

Never commit passwords or tokens.

Part E: Back to our results



Stage 0 set the goal. Stage 1 is done: data, fit and table are saved. Now: which figures carry the message?

Stage 0: One message steers every figure

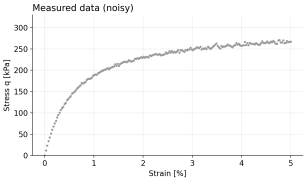
Working title - Message in one sentence

Identifying Duncan-Chang parameters from one triaxial test: manual guess versus optimizer

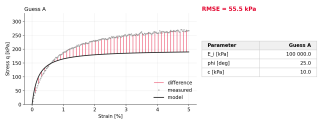
Every figure must support the message. If it does not, leave it out.

Stage 2 and 3: Five figures tell the story

Methods

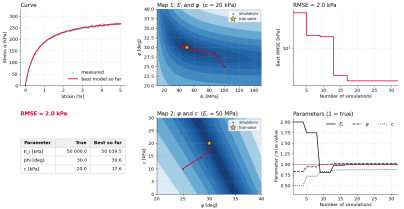


1 Input: noisy data

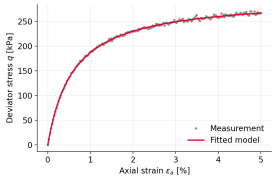


2 Judge: the RMSE

Results

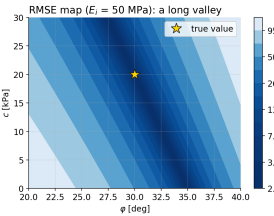


3 Search: the RMSE falls



4 Result: a good fit

Discussion

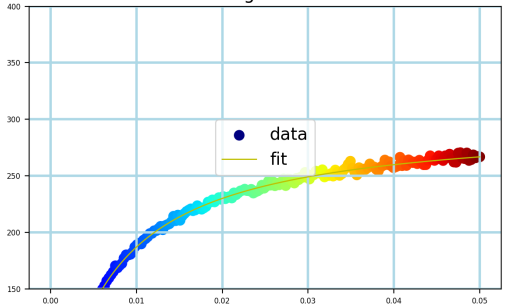


5 Limit: many pairs, one curve

Visualization: the same data, two figures

Bad

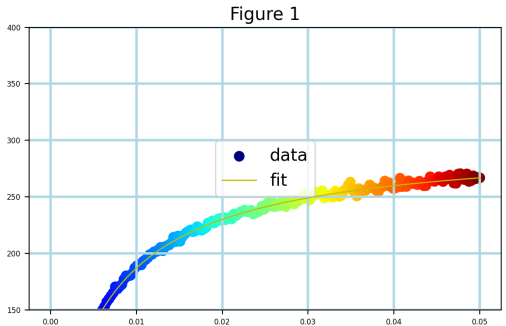
Figure 1



No labels, no units. Rainbow colours. Cut axis. Legend hides data.

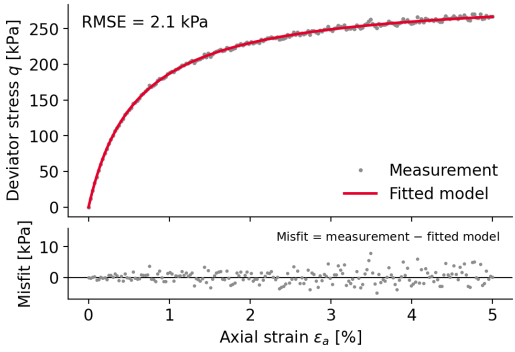
Visualization: the same data, two figures

Bad



No labels, no units. Rainbow colours. Cut axis. Legend hides data.

Good



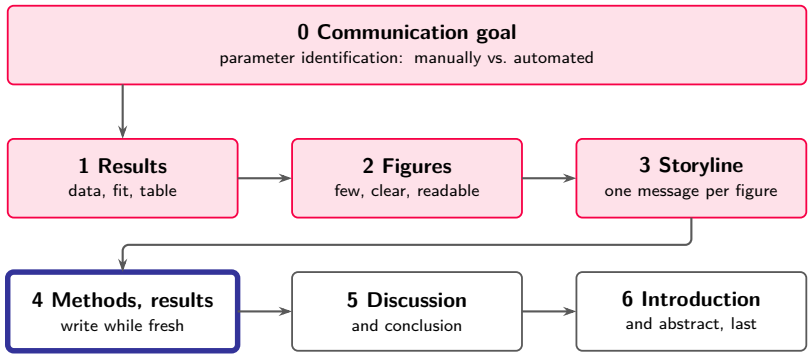
Labels with units. Data and model are clear. The misfit is shown.

Visualization: rules for a good figure

- One message per figure.
- Label every axis. Give the unit.
- Text in the figure is as large as the text of the paper.
- Use few colours. Avoid rainbow maps. Check for colour blindness.
- Show the data as points and the model as a line.
- Show the uncertainty or the misfit.
- Use the same style in all figures.
- Make every figure with a script. Save it in the repository.
- Make a clear legend. Avoid legends that hide the data.
- And clear figure captions that stand alone.
- **Discuss all figures in the main text (otherwise leave the figure out).**

Where are we? Stages 1 to 3 are done

A Warm-up | B Roadmap | C Live demo | D Repository | **E Paper** | F Outlook



We have results, figures and a story. Next: the structure of the text.

Structure: the shape of a paper (and a thesis)

Introduction: the problem and the research question

Background: what the reader must know

Methods: every step, so others can repeat it

Results: what we see

Discussion: what it means, and the limits

Conclusion: the answer to the question

Structure: what each part does in our example

Part	Job	In our demo
Introduction	Problem, gap, research question	Can we find E_i, φ', c' from one lab test?
Background	What the reader must know	The Duncan-Chang model and its three parameters (+ figure)
Methods	Every step and every input	1 % noise, $\sigma'_3 = 100$ kPa, least squares, seed 42, versions
Results	What we see. No opinion yet.	Fit, parameter table, RMSE map
Discussion	What it means. The limits.	E_i is found well. One test can identify only one combination of φ' and c' (**EXPLANATION!**). Data are synthetic.
Conclusion	The answer to the question	One test is not enough. More tests at other σ'_3 narrow the uncertainty of φ' and c'.

Write the text of a figure: describe first, then conclude

Bad

The results are shown in Fig. 3. It can be seen that the fit is very good, but φ and c are not good. This is because of the noise. The method is robust.

Write the text of a figure: describe first, then conclude

Bad

The results are shown in Fig. 3. It can be seen that the fit is very good, but φ and c are not good. This is because of the noise. The method is robust.

Good

Figure 3 shows the RMSE for pairs of φ and c. The minimum is a long, narrow valley. For φ from 24° to 35°, the best c falls from 43 to 0 kPa. The RMSE stays at 2.0 kPa. So with one test, many pairs of φ and c give the same curve. The optimization algorithm finds only one combination of many possible ones (e.g., the failure stress q_f). Tests at other cell pressures would give more combinations.

A caption stands alone

Bad

Figure 2. Result of the fit.

A caption stands alone

Bad

Figure 2. Result of the fit.

Good

Figure 2. Deviator stress q against axial strain ε_a for the synthetic triaxial test ($\sigma'_3 = 100$ kPa). Grey points: measurement. Red line: fitted Duncan-Chang model. Lower panel: misfit. RMSE $= 2.1$ kPa.

The reader should understand the figure without the text.

Write the method: every step, every input

Bad

We fitted the model to the data with an optimizer. The noise was small. The parameters are good.

Write the method: every step, every input

Bad

We fitted the model to the data with an optimizer. The noise was small. The parameters are good.

Good

We simulated one triaxial test with the Duncan-Chang model ($E_i = 50\,000$ kPa, $\varphi' = 30°$, $c' = 20$ kPa, $\sigma_3 = 100$ kPa, $R_f = 0.9$). We added 1 % Gaussian noise (seed 42). The seed sets the random numbers, so everyone gets the same noisy data. We fitted E_i, φ' and c' with bounded least squares (SciPy). The start values were 100 000 kPa, 25° and 10 kPa. Code, data and package versions are in the repository.

Write the discussion: explain, then give the limits

Bad

The results are good. The optimizer works well. The parameters φ' and c' are a bit uncertain. This can be improved in the future.

Write the discussion: explain, then give the limits

Bad

The results are good. The optimizer works well. The parameters φ' and c' are a bit uncertain. This can be improved in the future.

Good

The optimizer found E_i within 0.1 % of the true value. It also found φ' and c' close to the true values (30.6° and 17.5 kPa, true 30° and 20 kPa). But the answer is not unique. The curve depends on φ' and c' only through the failure stress q_f. One test at $\sigma'_3 = 100$ kPa gives one value of q_f. So many pairs fit equally well. Another start value can give another pair. Real tests behave the same way. More tests at other cell pressures give other values of q_f. In our demo, they narrowed the uncertainty of φ' and c'. Our limits: the data are synthetic, the noise is Gaussian, and we used one soil model.

Write the conclusion: answer the question

Our question was: can we find E_i, φ and c from one lab test?

Bad

In this study, a model was fitted. The results are interesting. More research is needed in many areas.

Write the conclusion: answer the question

Our question was: can we find E_i, φ and c from one lab test?

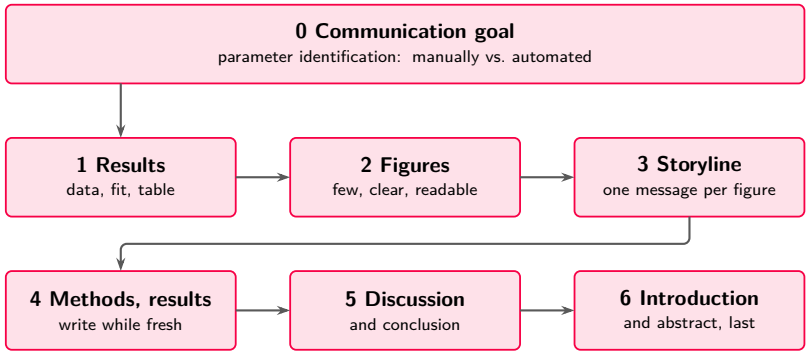
Bad

In this study, a model was fitted. The results are interesting. More research is needed in many areas.

Good

One lab test finds E_i well (fitted 50 040 kPa, true 50 000 kPa). With one test, many pairs of φ' and c' fit equally well. More tests at other cell pressures σ_3' narrow the uncertainty of both parameters.

Where are we? From results to paper



The order of writing is not the order of reading. Plan the paper from the results.

Practice: read a good paper in five minutes

1. Read the title and the abstract.
2. Look at the figures and tables.
3. Read the conclusion.

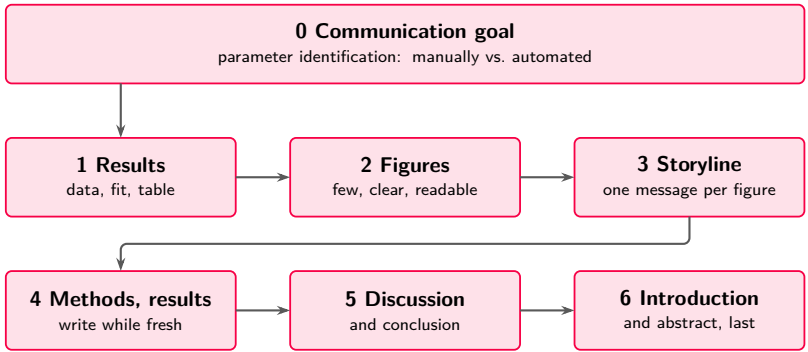
Then answer:

- What and where is the research question?
- Does the conclusion answer it?
- Could you repeat the method?
- Where are the data and the code?

Our paper (open access)

doi:10.1007/s11440-026-02951-5

Part F: Beyond the paper



The paper is done. A result can also become a tool that anyone can use.

Outlook: our ML model for the Hardening Soil model

- Same idea, applied to a research result.
- Two lab tests in, 12 soil parameters out.
- It runs in the browser. It is free.

Felić et al. (2026), Acta Geotechnica 21:4061–4078.
doi:10.1007/s11440-026-02951-5

Online tool



harifel.github.io/MLpFEM

Repository



github.com/harifel/MLpFEM

Take-home message

Coding is not a burden anymore, especially with coding agents.

- With a template and a clear structure, you start in minutes.
- A virtual environment makes your code run on every computer.
- A repository keeps your work safe and easy to share.
- Plan the paper from the results: figures first, introduction last.
- Use our template repository as your starting point.

Get the project and try it yourself

- The repository is public.
- It contains the slides, the notebook, the code and the data.
- Download it. Create the virtual environment. Run the notebook.
- Change the parameters. Try your own guess.

https://github.com/harifel/duncan-chang-calibration